

A Non-Visual Baseline for Multi-View Visual Question Answering on GoldenViewVQA

Aravind Mandiga

University of Georgia

aravind.mandiga@uga.edu

Abstract

We present a non-visual baseline for GoldenViewVQA, a benchmark for multi-view visual question answering with evidence-source identification. The baseline reaches a joint accuracy of 0.7119 on the hidden test split and uses no visual input. Its view prediction is the fixed label `CAM_FRONT`, which reaches 0.7797 view accuracy on the test split, while Qwen3-4B predicts the answer from the question, answer options, and non-visual metadata with 0.8475 answer accuracy. These results show that the current benchmark permits substantial performance without image access. We analyze the supporting-view prior and performance across question groups, and discuss benchmark changes that could reduce reliance on non-visual shortcuts. We present this run as a strong non-visual baseline, not as evidence that vision is unnecessary for driving question answering. Code: <https://github.com/aravind-mandiga/groundlm-goldenviewvqa>.

1 Introduction

GoldenViewVQA (Wang et al., 2026a) asks a system to answer a multiple-choice question about a driving scene and to say which of six synchronized nuScenes camera views (Caesar et al., 2020) holds the evidence for that answer. GoldenViewVQA is one of two GroundLM 2026 shared-task benchmarks, alongside LitTraceQA (Liu et al., 2026); the shared-task setup and findings are summarized by Wang et al. (2026b). The released data has 55 labelled development examples and 59 test examples without labels. Images are given as nuScenes file paths rather than distributed with the task.

We entered as team `aravind`. The non-visual baseline reported here predicts the supporting view with a fixed constant and predicts the answer with Qwen3-4B from non-visual inputs. It obtains 0.7119 joint accuracy and uses no image content.

Our contributions are:

1. We establish a strong non-visual baseline for GoldenViewVQA: a fixed view prediction combined with Qwen3-4B reaches 0.7119 joint accuracy on the hidden test split.
2. We quantify the supporting-view class prior and analyze answer and view accuracy across question groups, showing where the non-visual baseline is strong and where it is weaker.
3. Based on these results, we discuss concrete benchmark changes that could reduce reliance on non-visual shortcuts and place greater emphasis on visual grounding.

2 Related Work

Unimodal baselines in visual QA. The finding that visual question answering can be largely solved without the image is an old one. The first VQA benchmark (Antol et al., 2015) turned out to be answerable well above chance by models that saw only the question, because they could exploit answer priors. This led to VQA v2 (Goyal et al., 2017), where every question is paired with two images that have different answers, so a blind model cannot do well. Other work rebalanced binary questions (Zhang et al., 2016) or changed the answer distribution between train and test so that memorized priors actively hurt (Agrawal et al., 2018). Kervadec et al. (2021) argue that overall accuracy on these benchmarks mostly measures how well a model matches priors, and suggest reporting accuracy on rare question and answer combinations separately.

Our results indicate a related concern for GoldenViewVQA. We do not claim that the benchmark is broken. Instead, we measure a strong non-visual baseline so that future systems can be compared against the performance available without image content. Language-only baselines played a similar diagnostic role in earlier VQA benchmarks.

Question answering over driving scenes. Recent driving VQA benchmarks built on nuScenes (Caesar et al., 2020) include NuScenes-QA (Qian et al., 2024), which builds questions from scene graphs, and DriveLM (Sima et al., 2024), which links perception, prediction and planning questions into a graph. GoldenViewVQA differs by asking not just for an answer but also for the camera view that supports it, making evidence localization an explicit part of the task. Our analysis shows that the current supporting-view distribution contains a strong forward-camera prior.

3 The View Prior

The supporting view is heavily skewed. On the 55 labelled development examples, CAM_FRONT is the gold view 69.1% of the time, CAM_FRONT_LEFT 14.5%, and NONE_OF_THE_ABOVE 5.5%. No other view goes above 3.6%.

Always predicting CAM_FRONT therefore gives 0.6909 view accuracy on development and 0.7797 on the hidden test split.

The cost appears clearly in the macro metric. Our `view_macro_accuracy` is 0.1667, or 1/6, because a constant predictor can obtain recall only for one class. This contrast between micro and macro view accuracy shows that the dominant-view prior captures much of the micro view score while providing no coverage of the minority views.

4 Experimental Setup

Data. The task releases 55 labelled development examples and 59 test examples without labels. The two splits are disjoint by question and by normalized question text. Each example gives six camera views, a question, four options, and metadata covering the scene identifier, frame index, timestamp and question group. Gold labels give the supporting view and the correct option.

Visual inputs. The released records contain nuScenes-relative paths for six synchronized camera views. The baseline reported here does not access the corresponding image content. The view prediction is fixed to CAM_FRONT, and answer prediction uses only textual fields and non-visual metadata from the released records.

Task decomposition. Each prediction is a pair: a supporting view and an answer option. The two are produced independently. The view component

receives no question or option text, and the answer component receives no view information and produces no view estimate. Because the components share no state, they can be evaluated and replaced separately, which is what makes the Run 1 to Run 2 comparison in Table 1 an isolated substitution of the answer module.

View prediction. The view component is a constant predictor whose output is the mode of the empirical supporting-view distribution. We estimate that distribution once on the 55 labelled development examples, where CAM_FRONT is the most frequent supporting view at 69.1%, and then hold the prediction fixed. The predictor takes no input at inference time and therefore emits the same label for every test item. It is a deliberately simple baseline for the view component rather than an attempt at evidence-source identification, and it is the reason `view_macro_accuracy` is exactly 1/6.

Answer model and decoding. The answer component is Qwen3-4B (Yang et al., 2025), used as released, served with vLLM (Kwon et al., 2023) on a single GPU with a maximum sequence length of 8192 tokens and a memory-utilization target of 0.85. Each prompt is rendered through the model’s chat template with a generation prompt appended and thinking mode disabled; Section 10 describes the failures encountered when these settings were not used. Decoding is greedy, that is temperature 0, with a generation budget of 512 tokens, and no stochastic sampling is requested. Inference over all 59 test items finishes in under a minute. We perform no training or fine-tuning.

Prompt construction. The prompt contains the question, the four answer options with their labels, and the non-visual metadata carried in the released record: the question-group label, the scene identifier, and the timestamp. It closes with a development-time instruction to prefer a concrete, physically plausible option over an insufficient-evidence option when such an alternative is available. The full template is in Appendix A. We do not ablate these fields, so we cannot attribute the answer accuracy to any one of them.

Answer extraction. We obtain the answer by free-form generation followed by parsing, rather than by scoring the four options under the model and taking the most likely. The generative form produces one model completion per item and leaves

System	joint	view	macro	answer
Random answer	—	—	—	0.2500
Run 1: const. + const.	0.1525	0.7797	0.1667	0.1864
Run 2: const. + 4B text	0.7119	0.7797	0.1667	0.8475

Table 1: Official evaluator metrics on the hidden test split for the two submitted runs reported here. Joint accuracy is the primary metric. The fixed view prediction is unchanged between Runs 1 and 2.

the prompt unchanged, but it makes the prediction depend on an output parser and admits the failure mode described in Section 10. Generated text is mapped to a label by a fixed procedure applied in order. We first remove any `<think>` block, covering both a well-formed block and a leading unterminated one. We then scan the remaining text from left to right for single-character tokens matching `\b ( [A-Za-z] ) \b` and return the first whose uppercase form is a valid option label *for that question*; matching against the question’s own label set rather than a fixed alphabet prevents accepting a letter that names no option. If no such character is found, the system emits the first option label as a fallback. We log this event, because a fallback prediction is well formed and would otherwise be indistinguishable from a real one. It was never triggered: 0 of 59 test items and 0 of 55 development items required it. Each prediction is written as one JSON record pairing the fixed view label with the extracted answer letter.

5 Non-Visual Answer Prediction

Our answer module reads the question, the four option texts, and non-visual metadata (question group, scene identifier, and timestamp). It never receives image content.

Table 1 gives the results. Replacing the constant answer with the Qwen3-4B answer module raises answer accuracy from 0.1864 to 0.8475 and joint accuracy from 0.1525 to 0.7119. Development answer accuracy was 0.7818; the higher test value should be interpreted cautiously because both splits are small.

5.1 Reported runs

The results reported in this paper correspond specifically to our non-visual baseline submissions and are not intended to represent the team’s highest-scoring submission to the online evaluator. We report two submitted runs. Run 1 paired the constant view with a constant answer letter and scored

Question group	<i>n</i>	answer	view (const.)
causality	27	0.8889	0.7778
counterfactual	15	0.8000	0.6667
intent prediction	13	0.5385	0.5385

Table 2: Development accuracy by question group. Intent prediction has the lowest accuracy for the non-visual answer model, and it is also where the constant view prior is weakest.

joint 0.1525. Run 2 changed only the answer module and scored joint 0.7119. The component scores therefore show that answer prediction was the main bottleneck in Run 1. The view predictions are identical between the two runs, so this substitution increases answer accuracy by 0.6611 and joint accuracy by 0.5594.

6 Error Analysis

6.1 Where the Non-Visual Baseline Is Weaker

The task sorts questions into three reasoning types. Table 2 gives development accuracy for each, and the spread is wide.

Causality questions are answered at 0.8889 and counterfactual questions at 0.8000, while intent prediction drops to 0.5385. A plausible explanation is that causal and counterfactual questions often state more of the relevant situation in text, whereas intent prediction may depend more strongly on an agent’s location and current behavior in the visual scene. We do not test this mechanism directly, so the group-level result should be treated as diagnostic rather than causal evidence.

Intent prediction is also the group where the fixed-view prior is weakest, 0.5385 compared with 0.7778 for causality. The development results therefore suggest that groups with lower non-visual answer accuracy may also be less well served by the dominant forward-view prior. Larger splits are needed to determine whether this pattern is stable.

6.2 The model is not exploiting option position

A non-visual model can score well by exploiting answer-position imbalance. That does not appear to explain our development result. Gold answer positions are A:14, B:12, C:12, and D:17, while our predictions are A:15, B:12, C:12, and D:16. The gold distribution is relatively balanced, so a fixed-position strategy cannot account for the observed answer accuracy.

The strong result instead indicates that the non-visual inputs contain useful predictive cues. Our

prompt explicitly favors concrete and physically plausible alternatives over insufficient-evidence options. We do not ablate this instruction, the question-group label, scene identifier, or timestamp, so the current experiments cannot determine which non-visual cues contribute most.

6.3 Where the remaining gap is

Our joint accuracy of 0.7119 is bounded above by the fixed view accuracy of 0.7797. Even perfect answer prediction could therefore raise our joint score by at most 0.0678 without improving the view predictor. Further progress beyond that ceiling requires a better view model; our experiments do not establish whether that improvement must come from visual input or could partly come from other non-visual signals.

7 Benchmark Design Implications

The results suggest several ways to make performance less dependent on non-visual shortcuts.

Account for view imbalance. Always predicting CAM_FRONT reaches 0.7797 micro view accuracy but only 0.1667 macro view accuracy. Rebalancing the supporting-view distribution would directly reduce this prior. Alternatively, the benchmark could report a class-balanced grounding measure more prominently alongside joint accuracy, or consider a class-balanced joint measure that gives minority views comparable weight. This would preserve answer correctness while making success on dominant views less influential.

Construct distractors against non-visual baselines. Our answer model reaches 0.8475 without image content, showing that the current non-visual inputs contain substantial predictive signal. During dataset construction, candidate option sets could be screened with language-only models and revised when those models perform well. This would not guarantee visual grounding, but it would provide a direct check against the type of baseline reported here.

Examine the reasoning-type mixture. On development, intent-prediction questions have lower non-visual answer accuracy (0.5385) than causality (0.8889) or counterfactual questions (0.8000). A larger share of question types that remain difficult without images could reduce the strength of non-visual baselines. Because these numbers come

from only 13–27 examples per group, this should first be verified on a larger development set.

Publish a strong non-visual reference baseline.

Reporting a strong non-visual baseline alongside multimodal systems would make it easier to separate gains over simple guessing from gains beyond the information available without image content. Our 0.8475 non-visual answer accuracy provides one such reference point for the current task.

8 Reproducibility

The reported non-visual baseline requires no training. The view component is the fixed string CAM_FRONT. The answer component uses Qwen3-4B with greedy decoding, a 512-token generation budget, thinking mode disabled, and the prompt in Appendix A. On one NVIDIA L4-class GPU, inference over all 59 test items finishes in under a minute. Reproducing the reported predictions does not require access to the nuScenes images.

9 External Resources and Compute

We disclose the external resources used by the reported GoldenViewVQA system. The reported baseline uses the released GoldenViewVQA records and Qwen3-4B (Yang et al., 2025). Although the benchmark records reference nuScenes images (Caesar et al., 2020), the reported predictions do not access image content. We use vLLM (Kwon et al., 2023) for model serving and Hugging Face Transformers (Wolf et al., 2020) for tokenization and chat templating. We use no commercial APIs, closed-weight models, additional annotated datasets, synthetic training data, or fine-tuned checkpoints. All reported inference runs use a single NVIDIA L4-class GPU.

10 Implementation Lessons

Two inference settings caused silent failures during development. First, applying an instruction-tuned model without its chat template produced malformed or continuation-style outputs. Second, a very small generation budget with Qwen3 thinking mode enabled could exhaust the budget before an answer letter was produced. Applying the chat template, disabling thinking mode, and validating raw generations eliminated these failures. These checks are simple but important for shared-task pipelines because a syntactically valid fallback prediction can hide an inference error.

11 Conclusion

The non-visual baseline reported here reaches 0.7119 joint accuracy on GoldenViewVQA by combining a fixed supporting-view prediction with Qwen3-4B answer prediction from textual inputs and non-visual metadata. The result shows that the current benchmark permits substantial performance without image content. A dominant forward-view prior explains the strong micro view accuracy, while the high answer accuracy shows that non-visual inputs contain substantial predictive signal. Group-level results suggest that intent prediction is more difficult for this baseline, although the development split is small. Stronger non-visual baselines, class-balanced grounding evaluation, and distractor construction that explicitly tests language-only models could help future versions better isolate the contribution of visual grounding.

Ethics and Responsible Use

The system here answers questions about driving scenes without perceiving them. This is a statement about how the benchmark is built, and it should not be read as a claim about deployable driving software. A model that picks the most plausible-sounding description of a traffic situation has no perception of that situation, and putting such a component into a perception or planning stack would be unsafe. We report the result so that benchmark scores are not mistaken for grounding ability.

The benchmark references nuScenes data, which was collected on public streets and released under its authors’ stated terms. Our reported system does not access or redistribute the image content and performs no person-level analysis. Its task output is limited to a camera label and an option letter. The underlying language model carries whatever biases its pretraining gave it. Our prompt also tells the model to prefer specific options over hedged ones, which is tuned to this option set and would be wrong in any setting where declining to answer is the safe response.

Limitations

Our result is measured on a 59-item test split and a 55-item development split. Both are small enough that the accuracies carry real variance, and the 0.0656 difference between our development and test answer accuracy shows this directly. The view prior comes from 55 labelled examples and may

not hold on a larger or rebalanced split. The per-group numbers in Table 2 come from only 13 to 27 examples each, so they show trends rather than precise estimates.

We do not ablate the question-group label, scene identifier, timestamp, or the prompt instruction about insufficient-evidence options. We therefore cannot attribute the answer accuracy specifically to option wording or determine which non-visual cues matter most. We do not evaluate a vision-based counterpart to this baseline in this paper, so we cannot measure how much visual grounding would improve performance relative to other modeling choices. Finally, our fixed view predictor makes no real attempt at evidence-source identification; its `view_macro_accuracy` of 0.1667 reflects this limitation.

A Answer Prompt

The full prompt template, with substitutions in angle brackets:

```
A self-driving car is at an
intersection or road scene.
Answer the question using
careful physical and commonsense
reasoning about driving.

SCENE CONTEXT: <group> question,
scene <scene>, t=<time>s

QUESTION: <question>

OPTIONS:

<label>. <text> (×4)

One option is correct. Options
that claim there is not enough
evidence are usually wrong when
a concrete, specific alternative
is available. Prefer the option
that is specific and physically
plausible.

Reply with only the letter.
```

The scene-context line contains only non-visual metadata from the released record. We retain the question group, scene identifier, and timestamp in the prompt. The two sentences about insufficient-evidence options were added during development after the model frequently selected those options. They are a prompting heuristic for this benchmark, not a general rule about driving. We do not report ablations for these metadata fields or for the heuristic.

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